Blank / Junk Logic

As built on 2026-11-08. Code: `core-pipeline/modules/blank_junk/` (`method.py` the ladder, `stage.py` the short page and duplicates). Numbers: `resources/canon/blank_junk_context_canon.json`.

Each page gets one verdict: **Main**, **Blank**, or **Junk / subtype**. The subtypes are Cover Page, Cards, Letter/Fax, Record Request, Instructions and Others. There is no Junk / Invoice.

Two terms used throughout:

- **Electronic Signature** = kv_extract found a provider signature on the page (`provider_signature.csv`, `signature_present = y`). Words such as "signed" or "signature" in the text never count on their own.
- **Previous / next page** = the nearest page before / after that is **not blank** (no text, or no ink in stage 1). Blank pages in between are skipped.

Pass 1: on the Tesseract text, rules only

The first rule that matches decides.

#	Check	Verdict
1	No ink, stage 1 says blank, no text, or the word "blank" in < 30 characters	Blank
2	"Intentionally left blank" on a page of ≤ 25 words	Blank
3	A cover mark (Accept, Unaccept, Cover) on a page of < 5 words	Junk / Cover Page
4	< 15 characters	Junk / Others
—	Anything else	Main for now. Pass 2 decides

Pass 2: on the best Final OCR text

The first rule that matches decides.

#	Rule	Check	Verdict
1	Blank	The pass 1 rules 1–4 again	as in pass 1
2	Visual	A visual page under 20 words that is not a card	Main
3	Handwritten	More than 30% handwritten	Main
4	**Signature**	An Electronic Signature here, **and** none on the previous or next page, **and** the model's p(junk) < 0.95	Main
5	Card	An ID or insurance card. The model's p(keep) ≥ 0.99 keeps it	Junk / Cards
6	Fax	"Fax" in a section header, and rule 4 did not keep the page	Junk / Letter/Fax
7	Contents page	"Index" / "Contents" / "Table of Contents" as a heading, plus 3+ lines with page numbers	Junk / Others
8	**Short page**	Content ends in the top 20% of the page. See below	see below
9	Classifier	The model says JUNK (and p(keep) < 0.99)	Junk / its subtype
		The model says BLANK	Junk / Others
		Otherwise	Main

Rule 8: the short page

Rule 8 applies only to a page that got past rules 1–7, so a signed page kept by rule 4 is never junked here. The first check that matches decides.

#	Check	Verdict
a	**The signature carries on from the page before.** The previous page ends in its Electronic Signature (kv_extract places it in the footer, or its signature sentence is in that page's last paragraph). This page has the provider's name or credentials (kv_extract), the signer's name, credentials (MD, DO, NP…), or a date	Main
b	**"End of Report"** on the page	Junk / Cover Page
c	**Fewer than 5 distinct words** that are not on the previous or next page	Junk / Others
	…unless those words are only "Diagnosis" / "Visit Diagnosis"	Main
—	Otherwise	the classifier decides (rule 9)

An invoice page is never junked by the short page rule.

After the ladder: keeps (both passes)

Check	Verdict
A junk page with Lab, Reference Range and Value on it	Main (lab result)
A junk page with 2+ ICD-10 codes, or a Social History header	Main
An invoice with a money amount **and** a date	Main (page type Invoice / Invoice Page)
An invoice without both	Junk / Others
A card the model called an invoice	Junk / Cards

Duplicates (pass 2, after the verdict)

These checks compare only Main pages that are not junk:

- **Duplicate:** lengths within 10%, and either the shorter page is contained in the other, ≥ 98% of the words overlap, or fewer than 8 distinct words differ.
- **Partial:** 95–98% overlap, or fewer than 15 distinct words differ.

The later page is always the duplicate. Two pages are never matched when:

- their dates of service differ;
- a signed page (rule 4) is apart from its look-alike.

Later: continuity

A junk page inside a Progress Note is made Main again, **except**:

- blank, tiny and cover-mark pages;
- short-page junk;
- contents pages;
- cards.